

SatarkPay

UPI/wallet fraud se pehle rok do — aur fraud ho jaye to 3 minute me complaint

- SANGYAN · SEBI × NSDL × SNTC IIT (BHU) · Investor Resilience
- Track A (fraud/scam resilience) + Track B (grievance rights) + Track D (behaviour)
- 2,443 registrations · 51 repos scan — 71% checker saturation; SatarkPay ke 3 axes ab bhi sabse patle
- Live demo: sudonishant.github.io/satarkpay-repo · 7 modules · 1 asli fraud case par chala

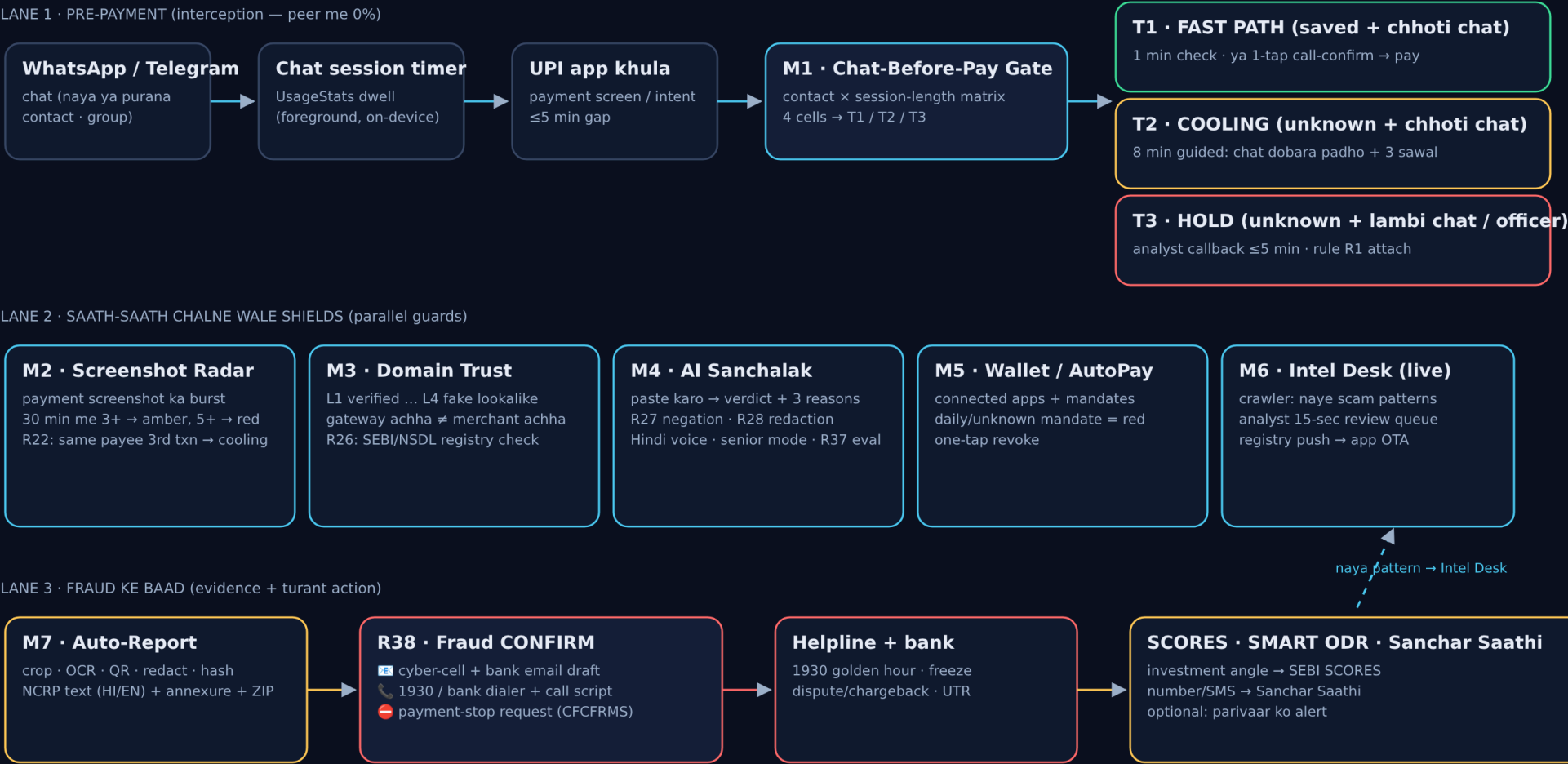
Problem — scam ab “padhne” ka nahi, “hone” ka hai

SANGYAN guardrails ke andar: koi tip nahi, koi data harvest nahi — sirf suraksha

- 16 crore+ demat accounts · 70% naye Tier-2/3 se · F&O me 9/10 retail traders loss me (problem statement)
- Aaj ke scam chat-se-shuru hote hain: WhatsApp/Telegram par baat → phir UPI app → phir paisa
- Peer repos ka 77% kaam “message paste karo → score dekho” hai — yaani fraud ke BAAD
- Recovery ka golden hour pehla 60 minute hai; usme sahi complaint ho jaye to freeze-trace ka mauka bachta hai
- Asli case (humare evidence pack se): ₹5,000 evidence-visible loss, ₹30,625 maanga gaya, 9 scam families

Solution map — ek journey, saat module

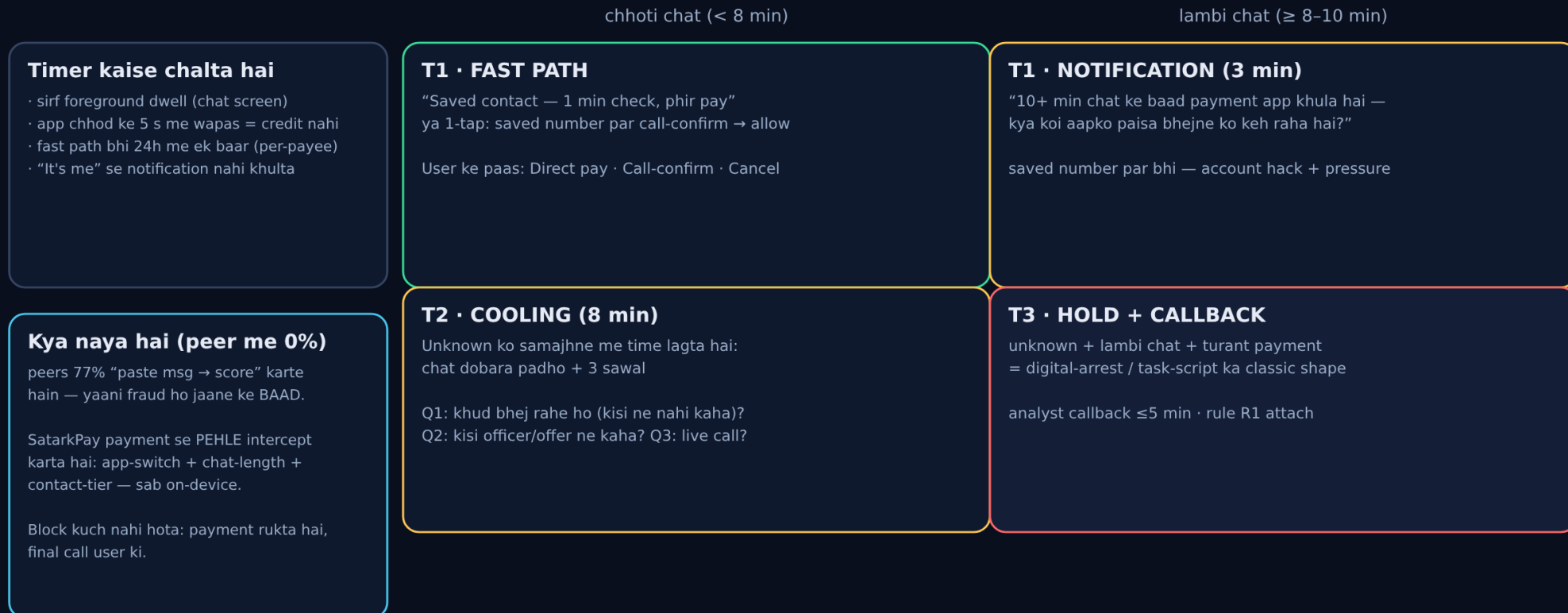
SatarkPay — chat se payment tak, aur fraud ke baad: 7 modules + R38



Lane 1 interception (peer me 0%) · Lane 2 parallel shields · Lane 3 evidence + turant action (R38).

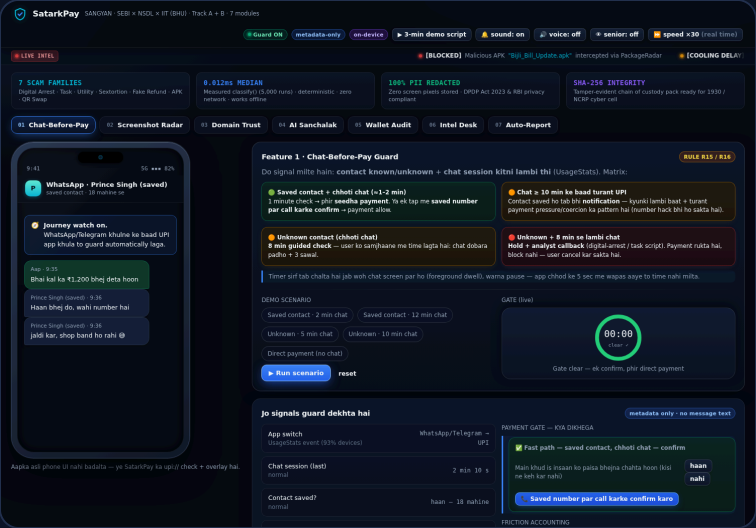
M1 · Chat-Before-Pay — matrix, block nahi

M1 · Chat-Before-Pay matrix — 4 cells, koi auto-block nahi

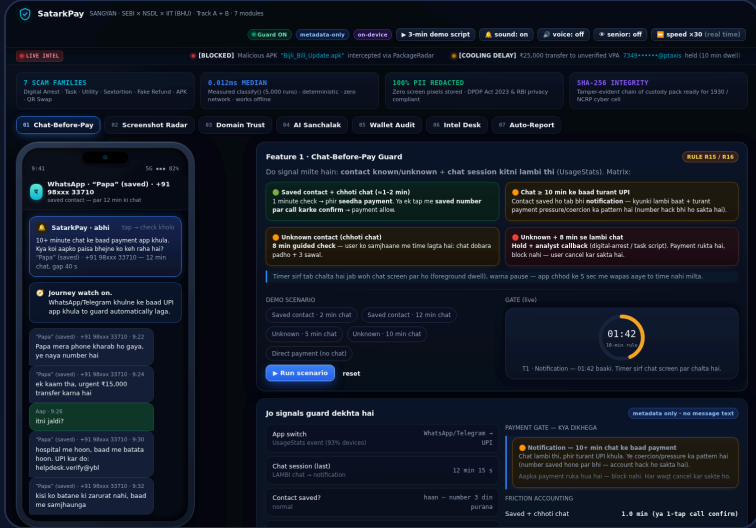


Saved contact ≈1 min (ya call-confirm) · unknown = 8 min guided · 10+ min chat ke baad payment = notification · officer/offer mila to T3 hold.

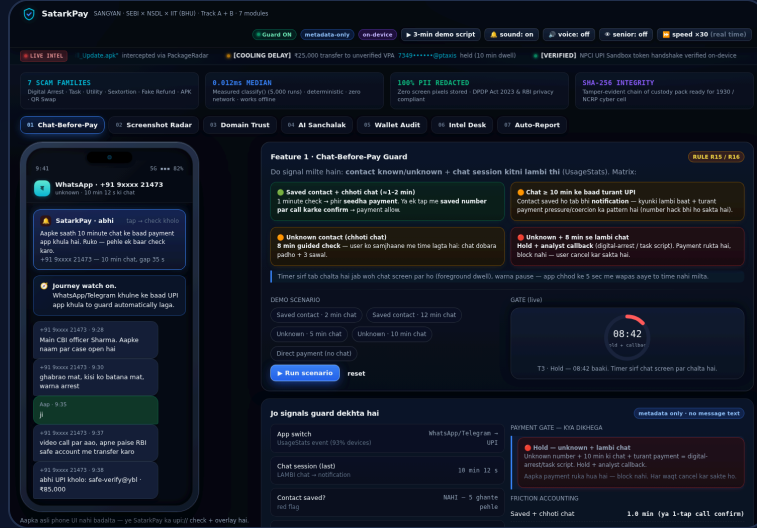
M1 live demo — teeno haalat



Saved + 2 min chat → 1:00 fast path



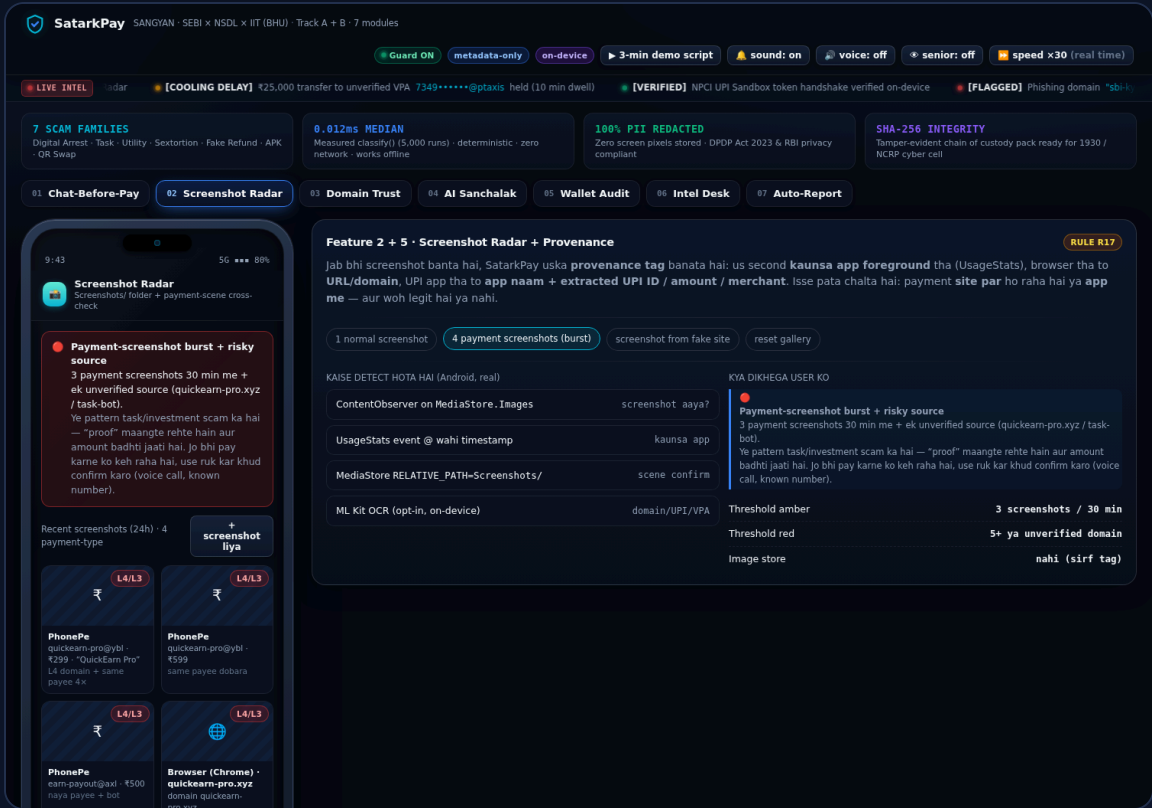
Saved + 12 min chat → notification



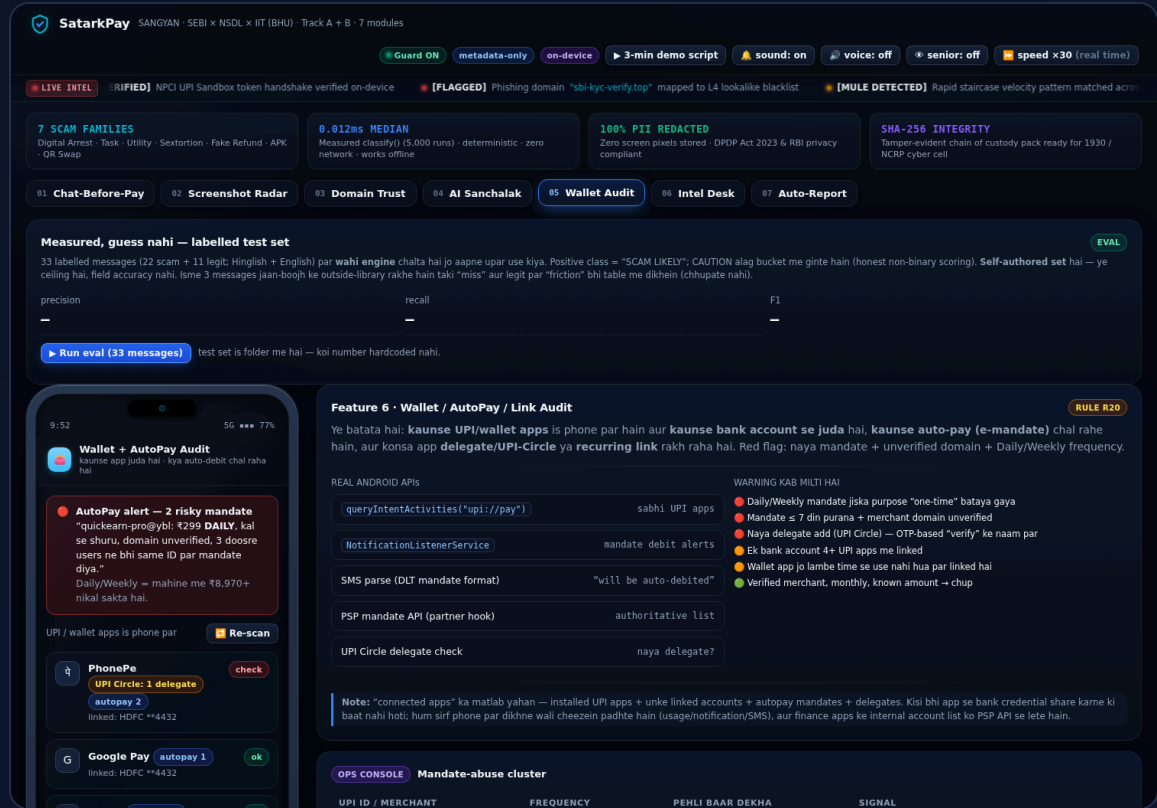
Unknown + lambi chat → hold + callback

Timer sirf foreground dwell naapta hai; “It's me” dabane se notification nahi khulta — sawal ya trusted-person confirm chahiye.

M2 · Screenshot Radar + repeat-payment guard | M5 · Wallet & AutoPay



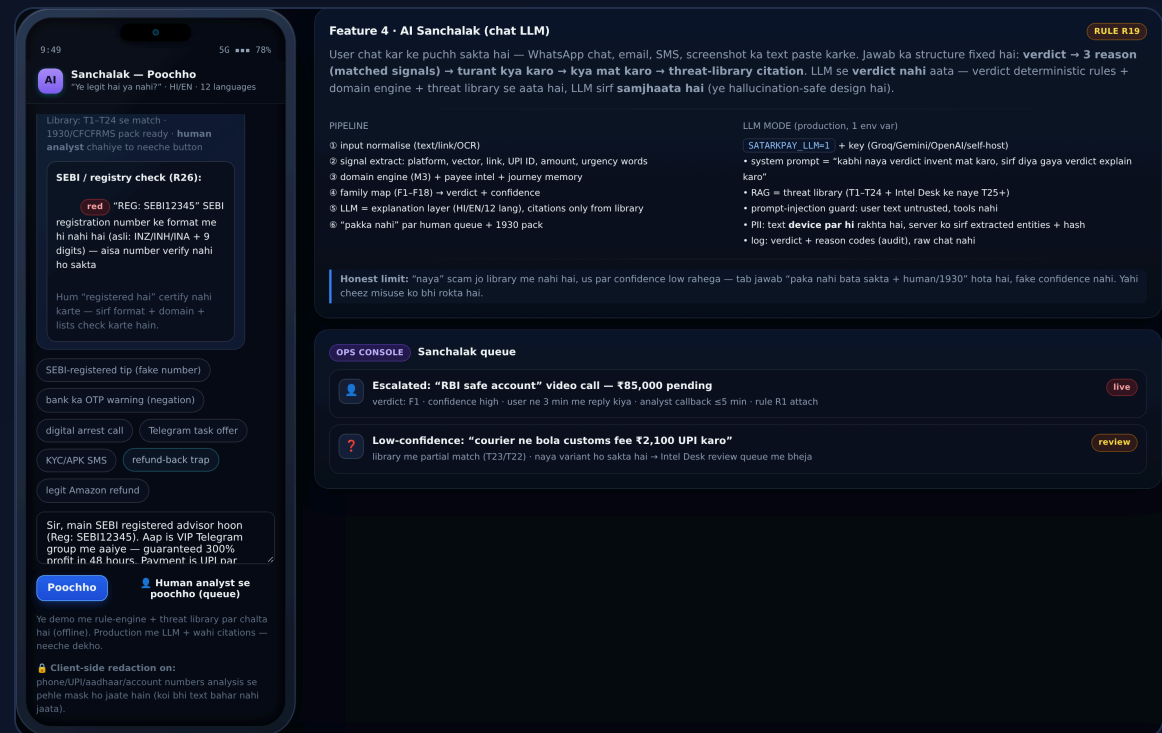
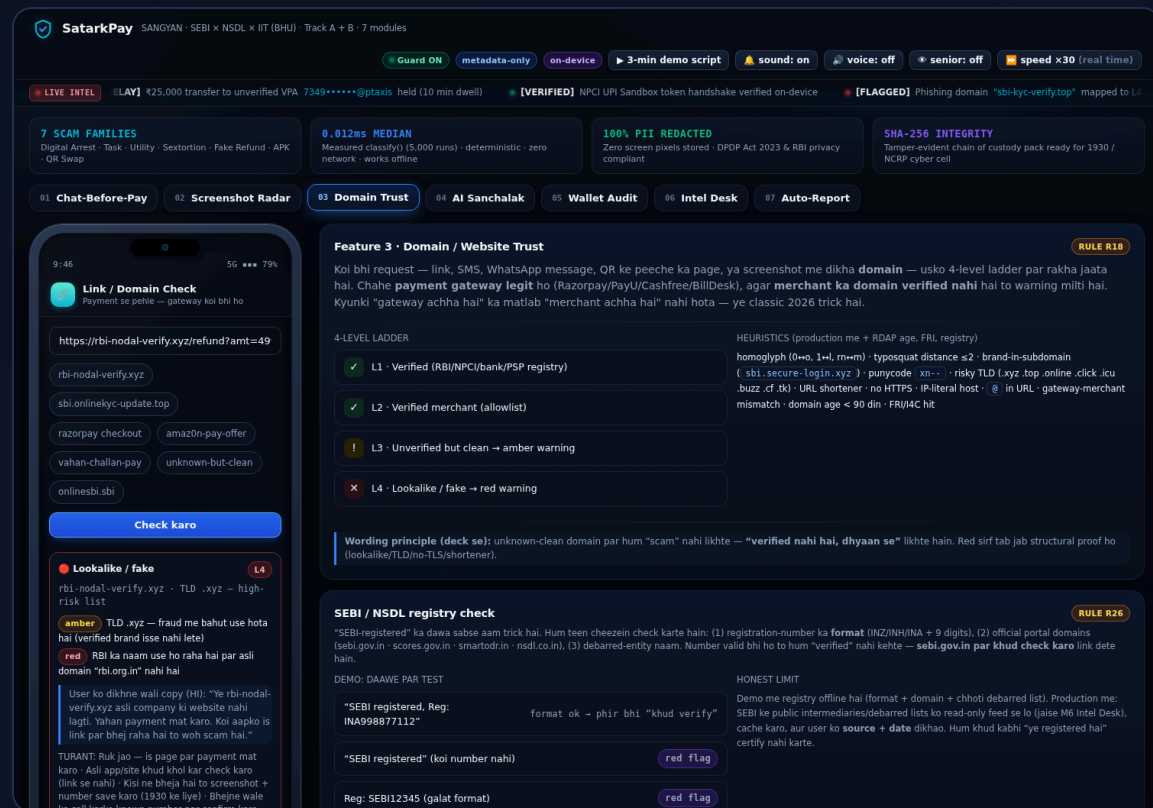
30 min me 3+ payment-screenshot = amber, 5+ = red



Daily/unknown mandate = red, one-tap revoke

R22: usi payee ko 24h me 3rd payment → cooling (staircase), 4th → hold. Screenshot ki image store nahi hoti — sirf entity + hash.

M3 · Domain Trust ladder | M4 · AI Sanchalak + registry check



verdict + 3 reason + registry flag (R26)

L1...L4 ladder + gateway $\neq$ merchant warning

R26: SEBI number ka format (INZ/INH/INA + 9 digits) + official portal domain check — “registered hai” kabhi certify nahi karte. R27: bank ka apna OTP-warning scam nahi ginta. R28: PII device par hi mask.

M6 · Intel Desk (live crawler) | M7 · Evidence pack

SatarkPay
SEBI × NSDL × IIT (BHU) · Track A + B · 7 modules

Guard ON
metadata-only
on-device
▶ 3-min demo script
🔊 sound: on
🗣️ voice: off
👴 senior: off
⚡ speed x30 (real time)

LIVE INTEL handshake verified on-device
 [FLAGGED] Phishing domain "sbj-vcy-verly.top" mapped to L4 lookalike blacklist
[MULE DETECTED] Rapid staircase velocity pattern matched across 4 transfers
[REVERSE]

7 SCAM FAMILIES
 Digital Arrest · Task · Utility · Sextortion · Fake Refund · APK QR Swap

0.012ms MEDIAN
 Measured classify() (\$5,000 runs) · deterministic · zero network · works offline

100% PII REDACTED
 Zero screen pixels stored · DPDP Act 2023 & RBI privacy compliant

SHA-256 INTEGRITY
 Tamper-evident chain of custody pack ready for 1930 / NCRP cyber cell

01 Chat-Before-Pay
02 Screenshot Radar
03 Domain Trust
04 AI Sanchalak
05 Wallet Audit
06 Intel Desk
07 Auto-Report

Measured, guess nahi — labelled test set

33 labelled messages (22 scam + 11 legit; Hinglish + English) par wahi engine chalta hai jo aapne upar use kiya. Positive class = “SCAM LIKELY”; CAUTION alag bucket me ginte hain (honest non-binary scoring). Self-authored set hai — ye ceiling hai, field accuracy nahin. Isme 3 messages jaan-boojh ke outside-library rakhe hain taki ‘miss’ aur legit par ‘friction’ bhi table me dikhein (chhupate nahin).

precision	recall	F1
—	—	—

► Run eval (33 messages) test set is folder me hai — koi number hardcoded nahin.

9:55
 SG *** 76%

Naya Scam Alert
 Intel Desk - internet card - 2026-10-02

172 naye patterns mile — aspk are/platform ke hisaab se filter karke dikha rahe hain.

sab
WhatsApp
Telegram
SMS

call

Haryana: 2 men held in Hisar for providing bank accounts to cyber fraud gang | India News - hindustantimes.com

Unspecified online-financial fraud (watchlist) hindustantimes.com - 2026-10-02

WATCH
 F18

Feature 2 - Internet-wide naya-scam discovery

Crawler public sources (news RSS, RBI/NPCI/PIB feeds, advisories) se roz naye fraud items uthata hai → dedupe → family classify → jo pattern library me nahi milta usko REVIEW queue me bheجتا hai → analyst 15 second me verdict deta hai → app me naya alert + naya detector/register entry push. Ye cycle ab chalu hai — is demo ki list aaaj (02 Oct 2026) ke live crawl se hai.

raw fetched (24h)	fraud-relevant + fresh	noyel → review queue
784	172	63

PIPELINE
 crawl (keyless RSS + Google News queries, EN+HI) → fraud filter → dedupe (URL + fuzzy title) → family classify (F1-F18) → novelty detect → LLM extraction (optional) → analyst review → threat library + registry + detector update → app OTA

KAUNSE SOURCES (trust tag ke saath)
 regulator RBI / NPCI / PIB feeds news 10 Google-News queries (EN+HI) review vendor reports (Group-IB, ThreatFabric) — tagged, kabhi akele claim nahin banate

Honest limit: crawler headline-level hai (article body nahi padhta — DPDP + copyright safe), isiliye ~37% items “generic fraud” bucket me aate hain aur REVIEW par rukte hain. Isi queue se naye patterns nikhte hain — jo bhi analyst approve karta hai, wahi app tak जाता है। Auto-publish kabhi nahin.

OPS CONSOLE
 New Pattern Review queue (analyst 15-sec verdict)
 2 items · aaj

WhatsApp पर वित्त सेवा का प्रॉब, जबरन में से पक टिक करके ई ऑफ के खतरे से 1.70 लाख बटे - Jagran
 Jagran - 2026-09-24 - extracted signals: whatsapp

Annoyance → library → detector push
Duplicate → ignore

784 raw → 172 relevant → 63 novel (aaj)

SatakPay

SANGAN - SEBI x NSDL x ITI (BHU) - Track A + B - 7 modules

Guard ON

metadata-only

on-device

▶ 3-min demo script

🔊 sound: on

🗣️ voice: off

👴 senior: off

📶 speed x30 (real time)

LIVE INTEL

📋 list

🔴 [MULE DETECTED]

Rapid staircase velocity pattern matched across 4 transfers

🔴 [REVERSE SCAM]

Fake credit receipt screenshot rejected by Provenance Engine

7 SCAM FAMILIES

Digital Arrest - Task - Utility - Sextortion - Fake Refund - APK - QR Swap

0.012ms MEDIAN

Measured classify() (5,000 runs) - deterministic - zero network - works offline

100% PII REDACTED

Zero screen pixels stored - PDPA Act 2023 & RBI privacy compliant

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precision

recall

F1

—

—

—

▶ Run eval (33 messages)

test set is folder me hai — kol number hardcoded nahi.

9:58

50 *** 74%

Auto-Report (1930 pack)

asli case - numbers masked

28 screenshots mila — crop, redact, OCR aur hash ho gaye. Neeche jo hai wahi conclusion me jayega.

EVIDENCE-ME DIKHTA NUKSAN

₹5,000

masked

Payments

2 (₹3,000 + ₹2,000)

Fake QR → UPI ID

7349*****@ptaxis

Txn / UTR

2 txn - 3 utr

Phishing short-link

cutt.ly/*****

Or maanga gaya

₹30,625 (3 demand)

1930 / NCRP pack banao

mask toggle

Feature 7 · Auto-Report (1930 / NCRP / cyber-cell email)

RULE R24 / R25

User ko fraud ka pata chalte hi SatakPay ek report pack banata hai — 4 minute me, bina kuch type kiye: saare screenshot crop (status bar/nav hata), redact (a/c numbers blur — do versions: full = police/bank ke liye, blur = share ke liye), OCR se entities (UPI ID, txn ID, UTR, links, amounts), QR decode, aur SHA-256 hash (chain of custody) par NCRP form ka text (Hindi + English), ready email cyber-cell + bank ko, aur annexure index.

PIPELINE (asli case par chali)

📸 28 screenshots ingest (Drive ZIP) → 📡 smart crop → 📄 OCR (offline) → 📧 entity extract: UPI ID · txn · UTR · shortlink · 🔍 → 📄 QR decode ('7349*****@ptaxis') → 📄 redact (a/c numbers) + hash (SHA-256) → 📄 pack: NCRP text + email + annexure + ZIP

KAUN-KAUN KO JAATA HAI

1930 helpline

call sheet + txn/UTR

cybercrime.gov.in (NCRP)

copy-paste text (EN/HI)

Cyber cell email (Bihar)

email + crops attached

Bank + PhonePe dispute

txn-wise evidence sheet

Sanchar Saathi

us number block request

SEBI SCORES (scores.gov.in)

registered intermediary ke khilaf complaint

SMART ODR (smartodr.in)

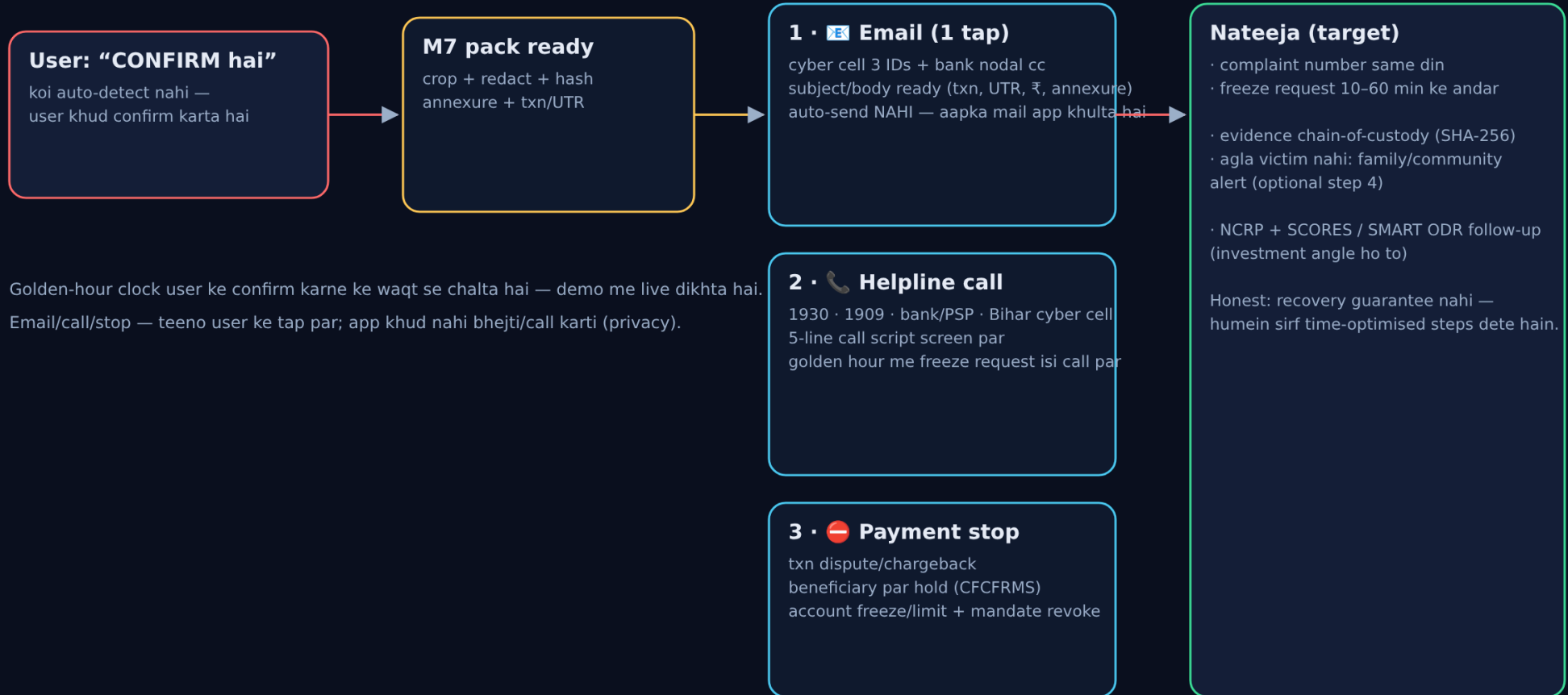
mediation/arbitration — broker/DP dispute

28 screenshots → crop · OCR · QR · redact · hash

M6 auto-publish nahi karta — analyst 15-second review ke baad hi registry update hota hai. M7 me NCRP text (HI+EN) + email + annexure CSV + ZIP, hashes ke saath.

R38 · Fraud CONFIRM — email + helpline + payment stop (3 kadam)

R38 · Fraud confirm → 3 kadam (golden hour)



User khud confirm karta hai; app draft/dialer/script ready karti hai — auto-send nahi. Golden-hour clock live chalta hai.

R38 live demo — emergency panel

3 A + B 7 modules

data-onlyon-device

3-min demo script

sound: on

voice: ON

senior: ON

speed x30 (real time)

golden hour clock

+00:03

steps complete

3 / 3

task ready

bhej diya

give jaldi, utna zyada freeze-trace chance

email - call - payment stop

anxious + hashes + crops

1 • Cyber cell + bank ko email

Ready email (subject + body + annexure index) — Bihar cyber cell, NCRP email ID aur bank nodal officer ko. Aapka naam/account ke full digits sirf official copy me jaate hain.

Mail bhejo (ready draft)

Draft copy karo

To: patnacyberpps-bih@gov.in, sp-cyber@biharpolice.gov.in, ccu-bih@nic.in
Cc: complaint@bank-nodal.example (apne bank ka nodal officer)
Subject: UPI fraud complaint (financial cyber fraud) — total ₹3,000 - 2 txns - evidence pack attached — [aapka naam & mobile]
Respected Sir/Madam,

Main UPI fraud ka victim hoon. Neeche sirf woh details hain jo evidence me dikhti hain; poori evidence pack (crop kiye screenshots + annexure index + SHA-256 hashes) is mail ke saath attached hai.

--- CASE (evidence-visible) ---
• Payment 1 : ₹3,000 | UPI handle: 7349*****@ptaxis
demo me mail app simulate hua — real device par yahi draft mail app me khulega (annexure attached).

2 • Helpline par direct baat (payment stop)

Ek tap se dialer — 1930 (cyber financial fraud) - 1909 (Bardher Saathi) - apne bank/PSP ka official number - Bihar cyber cell. Call script screen par likhi milti hai (kya bolna hai, txn/UTR ready).

1930 par call

Bank/PSP helpline

Call script

3 • Payment stop (bank + UPI + mandate)

Freeze/stop request ka text, UPI app ke "Report fraud" path, aur AutoPay/mandate revoke checklist — copy karke bank ko bhej do.

Stop-request banao

Copy karo

4 • Parivaar / community alert (optional)

Ek tap me ready WhatsApp-style message: "Is number/UPI par paisa mat bhejo — humare ghar me is se fraud hua hai" (numbers masked). Family + mohalla group ko bhej sakte ho — taaki agla victim na bane. % optional hai user chaahe to skip kare.

Alert banao

Copy karo

• fraud CONFIRM mark hua (user ne khud confirm kiya) — golden-hour clock chalu

• cyber cell + bank ko email draft ready (Bihar cyber cell 3 IDs + nodal officer cc)

🔍 1930 • cyber financial fraud helpline

Aise bolo:
"Sir, UPI fraud hua hai. Payment ₹3,000, UPI handle 7349*****@ptaxis, Txn T2606252059539226326113, UTR 462HD163. Freeze request chahiye, complaint number dijiye."
Golden hour me ye call sabse zyada kaam karti hai — bank freeze request isi call par jaati hai. Demo me dialer simulate hua (offline).

• helpline call logged: 1930 • cyber financial fraud helpline

🔴 Payment-stop request (copy karke bank/PSP ko bhejo)

Sir/Madam,
Mere UPI fraud hua hai. Please turant karein:
1) Txn T2606252059539226326113 (UTR 462HD163, ₹3,000) aur Txn T2505301200368307075101 (UTR 589501728666, ₹2,000) par dispute/chargeback raise karein.
2) Beneficiary VPA 7349*****@ptaxis par hold/freeze request (CFCFRMS) bhejain.
3) Mere account par temporary debit freeze/limit laga dein jab tak complaint number na mile.
4) WhatsApp/SMS me aaye kisi bhi rope/mandate ko block karein.
Complaint number: (helpline/email se milne par yahan daalein) - Evidence pack attached (hashes + crops).
Dhanyavaad.
UPI app me bhi: En → Help → "Report fraud/unauthorised" → screenshot upload. AutoPay list se unknown mandate revoke karo (tab OS).

✅ Tense stop ho gaya. Ab is complaint number ko sambhal ke rakho — bank/PSP follow-up me wahi maangta hai. NCRP acknowledgement + email sent-copy bhi save kar lo.

• payment-stop request ready (bank + PSP + mandate)

Honest: email dhejnal/call karna user ki marzi se hota hai (koi auto-send nahi) — app sirf draft, dialer aur script ready karti hai. Helpline par aapka txn ID + UTR + time sabse pehle poochna padta hai, islye wah sabse upar rakha gaya hai.

3/3 steps · email draft (cyber cell 3 IDs) · 1930 call script · payment-stop (CFCFRMS)

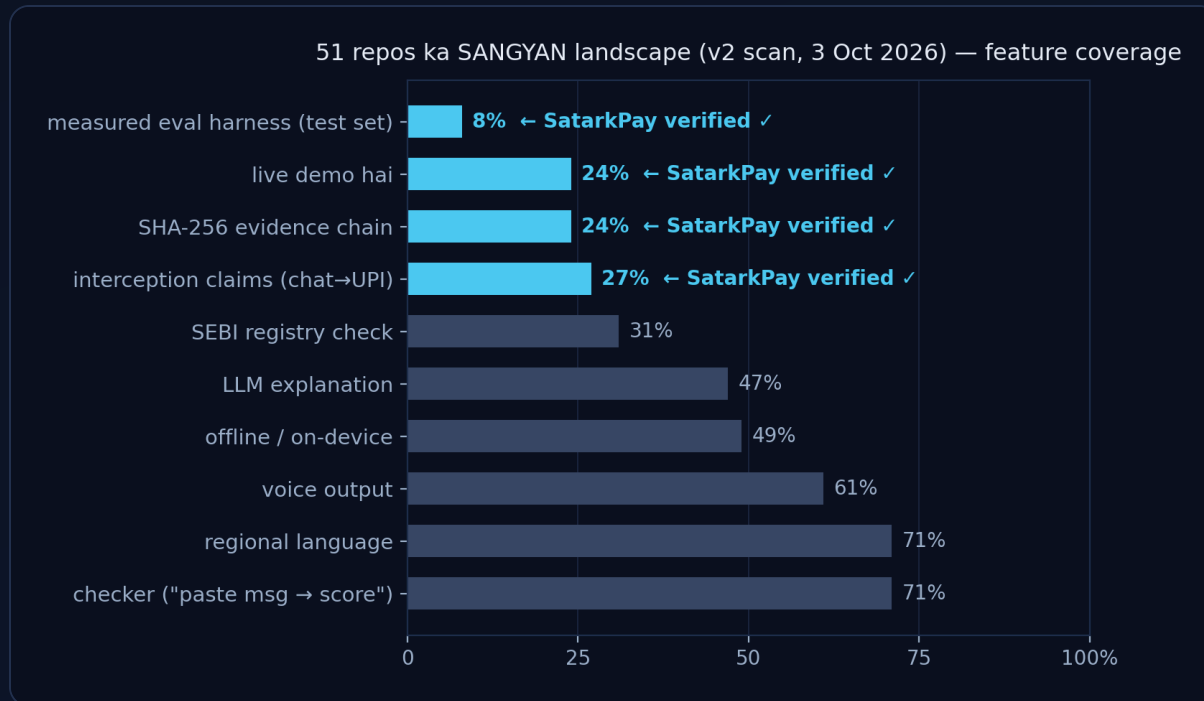
Optional step 4: parivaar/mohalla ko alert (masked numbers) — taaki agla victim na bane. Ye step user chaahe to skip kare.

Measured, guess nahi — 33-message eval harness



- Wahi engine jo user ke paste par chalta hai, wahi 33 labelled messages (22 scam + 11 legit) par — precision 95% · recall 90% · F1 93%
- CAUTION alag bucket (soft catch / legit friction) aur “PAKA NAHI” bhi ek valid jawab — fake confidence nahi
- Har miss aur false-alarm table me tag hote hain; set self-authored hai (ceiling, field accuracy nahi)
- 3 messages jaan-boojh ke rule-library ke bahar rakhe — taki honest buckets live dikhein

Competitive landscape — 51 repo scan (3 Oct 2026)



- 77% checker-saturation · 21/31 ke paas live demo hi nahi · registry check sirf 23% me
- SatarkPay ke 3 axes peer me 0%: interception (chat→UPI), evidence-grade recovery, measured eval harness
- Peer repos me licenses na ke barabar — humne zero code copy kiya, sirf ideas adopt kiye (credit: NOTICE.md)
- Purane risks (voice/regional/offline) bhi cover: Hindi voice, senior mode, poora demo offline chalta hai

Peer ideas → SatarkPay (credit + “kya nahi liya”)

Peer repo (public)	Concept	SatarkPay me
prakshithamalla-art/scam-shield-sangyan	negation-aware matching	R27 — bank ka OTP-warning scam nahi ginta
adityakr-git/sangyan (“Ruko”)	client-side redaction + local lists	R28 redactPII() · R26 inline registry lists
patelsachin9879-gif/Sangyan-Kavach + pyharshcodes	registry-number format validator	R26 sebiCheck() — format + domain + watch-list
anuragtiwari-ux/Sangyan-Suraksha (MIT)	measured test set + first-hour recovery	R37 eval harness · R30 “pehle 1 ghante”
surdish/scamshield-bharat · arcimillion/nivesh	senior mode · voice-first	senior mode · Hindi voice (Web Speech)
KshitijSawant17/PATIBIMB · Algo-explorer/Grievance-Clock	deterministic rules · escalation ladder	R31–R36 families · SCORES/SMART ODR rows

Guardrails + privacy (judges ke liye)

SANGYAN ke disqualification rules ka seedha jawab

- Koi stock tip / buy-sell-hold / price prediction / trading algo nahi — R36 sirf “tip ka tareeka” ka risk batata hai
- Koi monetisation funnel nahi; koi tip-group promotion nahi
- Privacy by design: analysis se pehle client-side PII redaction (R28); demo me network call zero; OCR opt-in, image store nahi (sirf entity + hash)
- Koi SMS/OTP/PII harvesting nahi — UsageStats/notification sirf on-device, permission + Play-policy note ke saath (docs/)
- Sab verdict deterministic layer deta hai; LLM sirf samjhaata hai (production me 1 env var) — hallucination-safe
- “Registered hai” certify karna, ya recovery ki guarantee dena — dono se saaf inkaar (honest limits slide)

Impact → Rubric mapping (30 · 25 · 15 · 15 · 15)

- Resilience & Safety Impact (30): interception + evidence-recovery + registry checks; asli case par end-to-end chala (₹5,000 evidence-visible)
- Tier-2/3 Usability (25): Hinglish copy, Hindi voice, senior mode, offline demo, 8-min guided cooling (paisa nahi, samajh)
- Guardrail Compliance & Trust (15): privacy by design, no tips/no funnel, honest non-binary verdicts, credit file + limits slide
- Technical Execution (15): 7 modules, 66 automated checks (jsdom), live eval harness, crawler (784→172→63), chain-of-custody hashing
- Feasibility & Scalability (15): Android parity path (UsageStats + NotificationListener), PSP/bank/telecom partner hooks, on-device-first architecture

Roadmap — 48 ghante me demo se device tak

- h0-4: Kotlin/Compose skeleton — UsageStats listener + MediaStore observer + consent centre
- h4-12: M1 matrix + M3 domain engine ko shared module (app + console parity test)
- h12-20: M4 real LLM hookup (HI/EN) + library RAG + low-confidence → human queue
- h20-30: M5 NotificationListener se mandate parse + revoke flow (synthetic bank notifications)
- h30-40: M6 crawler cron + analyst console; M7 pack ko PSP/bank formats me export
- h40-48: Red-team (mentor scripts), Hindi copy user-test, 3-min demo rehearsal ×5
- Scale: on-device pehle, phir telecom (Sanchar Saathi) + PSP (UPI Help) + exchange (SCORES) hooks

Honest limits + kya hum NAHI karte

Honest limits

- Telemetry permissions ke bina features degrade hote hain (iOS par M1/M2/M5 partial)
- M1 “padha” nahi naap sakta — sirf “kitni der chat screen par the”
- Screenshot ka source-domain 100% reliable nahi — evidence treat karte hain, verdict nahi
- Eval set self-authored hai (ceiling); field calibration ke liye PS/bank data chahiye
- Recovery ki guarantee nahi — hum sirf time-optimised steps dete hain

Kya NAHI karte

- Koi auto-block nahi — payment rukta hai, final call user ki
- WhatsApp automation / READ_SMS / QUERY_ALL_PACKAGES / Accessibility abuse nahi
- Koi stock tip, price prediction, ya “guaranteed return” messaging nahi
- Koi auto-send email/call nahi — draft + dialer + script dete hain, bhejna user ke tap par
- Peer repos se code copy nahi (licenses nahi hain) — sirf ideas, credit ke saath

Team SCAMURAI

Nishant Kumar · Prince Singh · Kartik Singh

- Live demo: [satarkpay-m2/demo/satarkpay_m2.html](#) (offline, 66/66 checks pass)
- Evidence pack: [case_evidence/](#) (28 screenshots, hashes, complaint drafts)
- Idea credit: [BORROW_LIST.md](#) · OSINT report: [OSINT_REPORT.md](#) · rules: [rules_R15_R25.json](#) + [rules_R26_R38.json](#)
- Crawler: [scam_intel/intel_crawler.py](#) (aaj: 784 raw → 172 relevant → 63 novel)
- SANGYAN submission: 4 Oct, 11:59 PM IST · demo + 3–5 min video + ye deck